

Iqbal Maqbool Sofi

Glassboro, NJ | iqbalmaqbol@gmail.com | [GitHub](#) | [LinkedIn](#) | [Research Portfolio](#)

Profile

M.S. Data Science graduate with industry experience in analytics and software engineering and research experience in deep learning, transformers, semantic communication, 3D representations, and generative reconstruction. Experienced in PyTorch-based experimentation, research-code debugging, and model evaluation under severe information loss.

Education

Rowan UniversityGlassboro, NJM.S. in Data Science2024 – 2026

Bachelor of Technology in Computer Science and EngineeringIndia

Research Experience

Semantic Communication for 3D Point CloudsRowan UniversityMachine Learning Research2026

- Investigated extension of the **Discernment semantic communication framework** to ModelNet40 3D point clouds represented by **4,096 discrete MaskedVQVAE3D tokens**.
- Implemented a **MONAI-style decoder-only transformer (DoT)** and compared it with the existing NanoGPT implementation under **50%–99.2% token loss**.
- Identified a transformer output-head dimensionality issue and traced anomalous ~2% classifier performance to an incorrect token dataset, restoring evaluation on the correct data.
- Obtained nearly identical validation losses to NanoGPT, while conditional NanoGPT retained a **6–12.5 percentage-point accuracy advantage** over MONAI at multiple severe corruption levels.
- Verified the token-to-geometry decoding path and identified an evaluation limitation involving class-sorted 200-sample subsampling versus the **67.9% full-validation clean baseline**.

Selected Projects

AudioMNIST VQ-VAE + TransformerPyTorch, MONAI, Transformers

- Built and evaluated a VQ-VAE + transformer pipeline for spoken-digit representations; trained a classifier reaching approximately **98.5% test accuracy** and evaluated robustness under FGSM adversarial perturbations.

Trust-Aware Machine Learning PipelinePython, ML Monitoring

- Developed a trust-scoring pipeline on **541K+ retail records**, detecting missingness and distribution shifts and producing interpretable data-reliability diagnostics.

Professional Experience

AccentureIndiaCustom Software Engineering AnalystFeb 2022 – Aug 2024

- Implemented enterprise analytics and data-tracking solutions using **Adobe Analytics, Adobe Experience Platform, Power BI, and Excel**.
- Supported analytics implementations across **80+ pharmaceutical websites** and a major UK banking client, including tracking validation, data governance, quality checks, and reporting.

DXC TechnologyIndiaAssociate Software DeveloperAug 2021 – Feb 2022

- Developed enterprise applications using **C#, .NET, SQL Server, and REST APIs**; contributed to debugging, API testing, and Agile sprint delivery.

Technical Skills

ML / Research: PyTorch, Deep Learning, Transformers, Vision Transformers, VQ-VAE, Generative Modeling, Semantic Communication, 3D Point Clouds, Model Evaluation

Programming / Data: Python, SQL, C#, .NET, Pandas, NumPy, MongoDB, InfluxDB

Tools / Platforms: Git, Linux, CUDA, Jupyter, Power BI, Adobe Analytics, Adobe Experience Platform